

Supporting Information

Synthesis of Aluminum Metal-Organic Frameworks

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About this sample

This document provides sample text for a text-mining demonstration. All quantities, conditions, and outcomes are supplied for demonstration purposes. Characterization descriptions draw on existing literature.¹ Detailed synthesis procedures appear in the Methods section of the main article.

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Characterization overview

MOF-303 and CAU-23 were examined by diffraction and sorption measurements to assess crystalline order and accessible porosity. Thermal analysis and electron microscopy provided complementary views of the recovered powders. Figures S1-S4 reserve space for the corresponding characterization panels.

Section S1. Powder X-ray diffraction

Powder X-ray diffraction (PXRD) used a Bruker D8 Advance or Rigaku Miniflex 600 with a copper anode and nickel filter. Measurements used Cu K α radiation in Bragg-Brentano geometry. Powders were placed on zero-background holders and leveled before measurement.

Patterns were collected over 3-50° in 2303 steps, corresponding to approximately 0.02° per step. The acquisition time ranged from 0.5 to 10 s per step. These settings provide a common basis for comparing crystalline order in MOF-303 and CAU-23.

Temperature-dependent crystallinity

Products prepared at reaction temperatures below 100 °C had no crystallinity, as confirmed by PXRD (Figure S1). This comparison concerns crystalline order in the recovered powder. The synthesis and workup conditions are described in the Methods section of the main article.

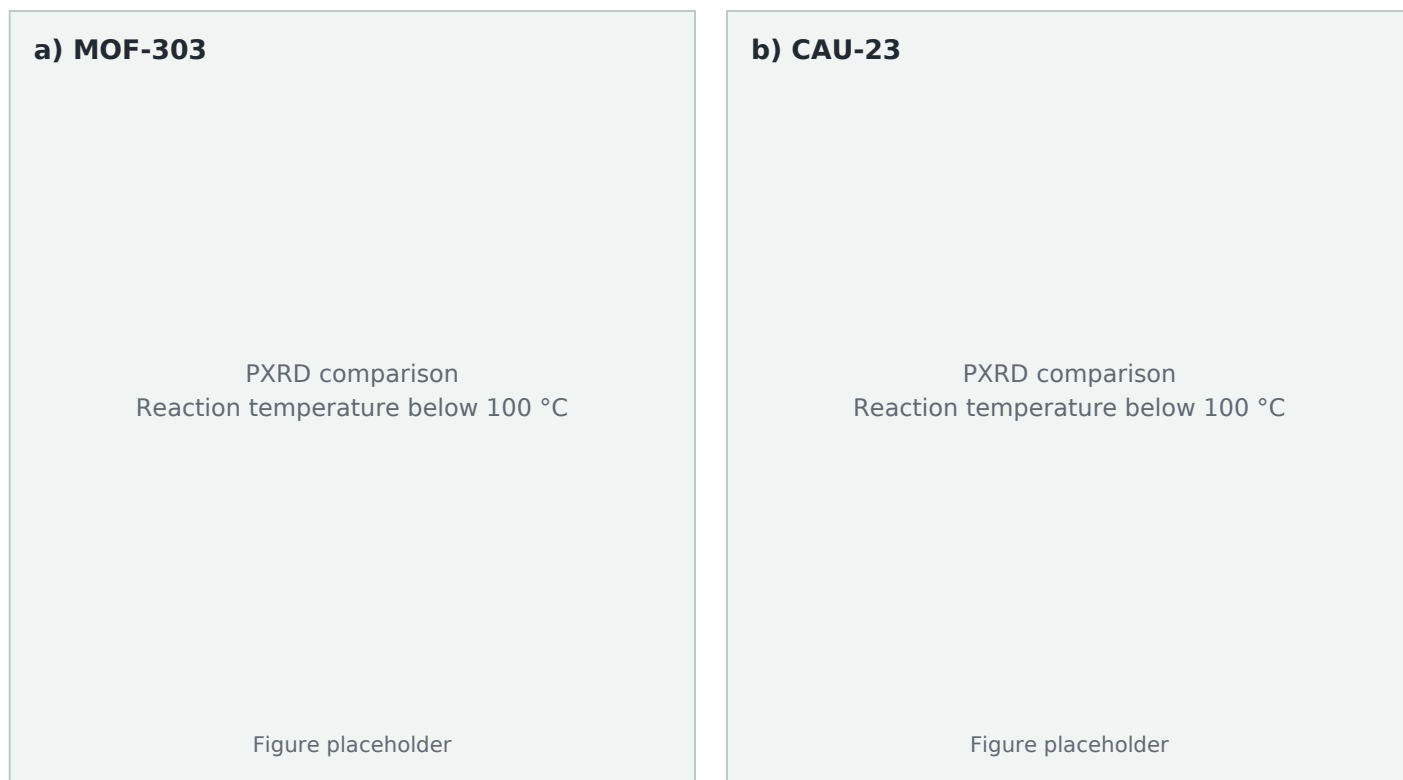


Figure S1. PXRD comparison for products prepared at reaction temperatures below 100 °C: (a) MOF-303 and (b) CAU-23. The panels are figure placeholders and contain no measured data.

Section S2. Sorption measurements

Nitrogen adsorption

Nitrogen sorption was measured at 77 K using a Micromeritics 3Flex or ASAP 2420. Ultrahigh-purity nitrogen served as the adsorbate, and helium was used for free-space corrections. The sample was cooled in a liquid-nitrogen bath. Data analysis used MicroActive software.²

Water vapor adsorption

Water sorption measurements used a BEL Japan BELSORP-aqua3. The water source was degassed through five freeze-pump-thaw cycles, and helium was used for free-space corrections. An isothermal bath controlled the sample temperature. Isotherms at 15, 25, and 35 °C were used to examine the temperature dependence of adsorption and estimate adsorption enthalpies.^{3,4}



Figure S2. Nitrogen sorption at 77 K for (a) MOF-303 and (b) CAU-23. The panels are figure placeholders and contain no measured data.

Section S3. Thermogravimetric analysis

Thermogravimetric analysis (TGA) used a Netzsch Jupiter STA449 F5. Before the measurement, samples were dried by heating to 150 °C at 1 °C per minute. The furnace was then allowed to cool to 40 °C.

For the analysis, the temperature was increased from 40 to 800 °C at 1 °C per minute. Ultrahigh-purity argon at 60 mL per minute supplied the balance purge. The sample purge used either argon for inert conditions or ultra-zero-grade air for oxidative conditions, also at 60 mL per minute.

The thermal comparison complements diffraction and sorption measurements by following mass changes with temperature (Figure S3).

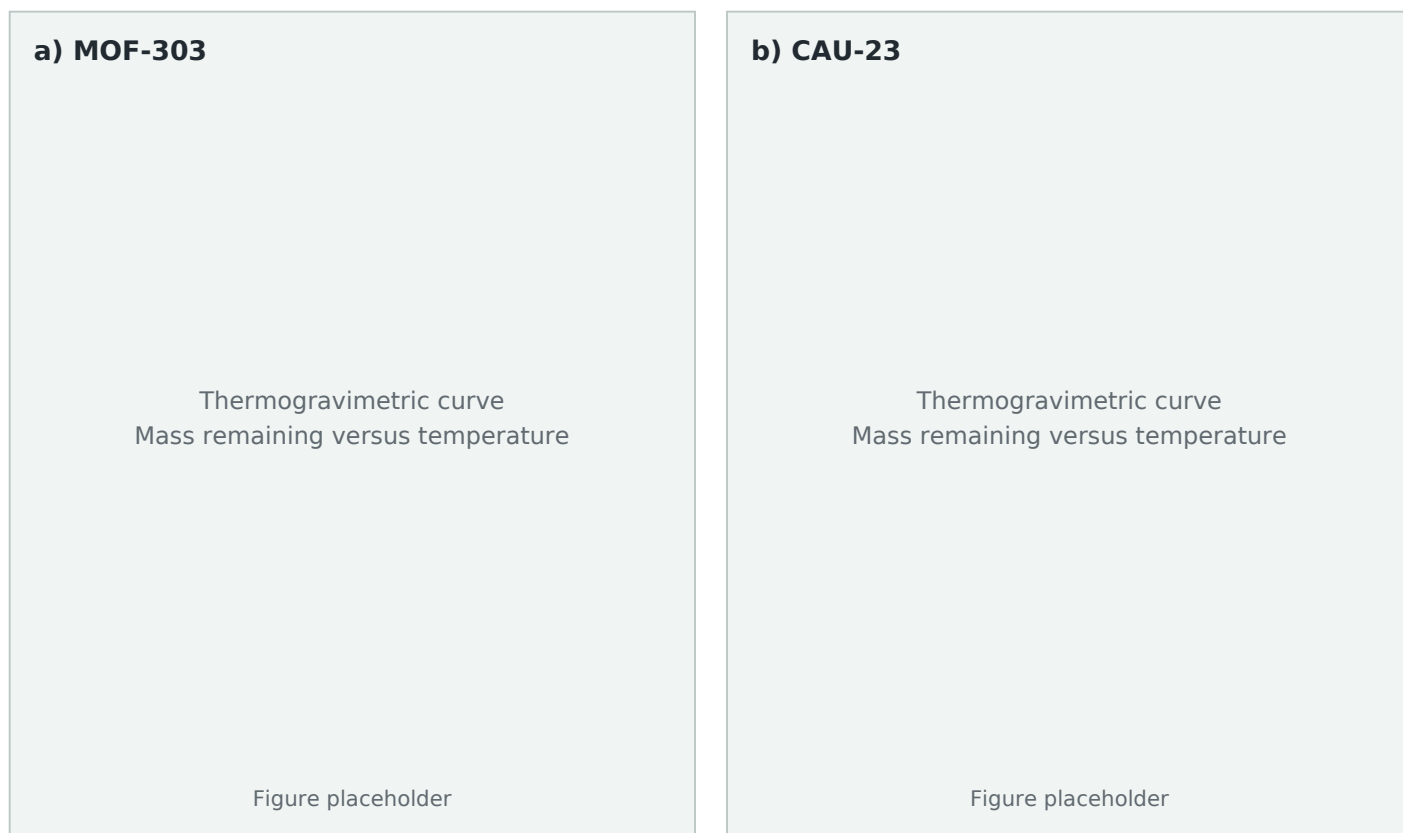


Figure S3. Thermogravimetric analysis of (a) MOF-303 and (b) CAU-23. The panels are figure placeholders and contain no measured data.

Section S4. Electron microscopy

Scanning electron microscopy (SEM) used a FEI Quanta 3D FEG at an accelerating voltage of 10 kV and a working distance of 10.0 mm. Energy-dispersive X-ray spectroscopy (EDS) used an Oxford X-Max system at 10 kV.

Powders washed with water and methanol were dispersed in ethanol and sonicated for 60 s to disperse larger agglomerates. A drop of the suspension was placed on a silicon wafer. After the ethanol evaporated, the deposited powder was dried overnight under dynamic vacuum (approximately 10^{-3} mbar). SEM was used to examine particle morphology, with EDS providing complementary elemental information.

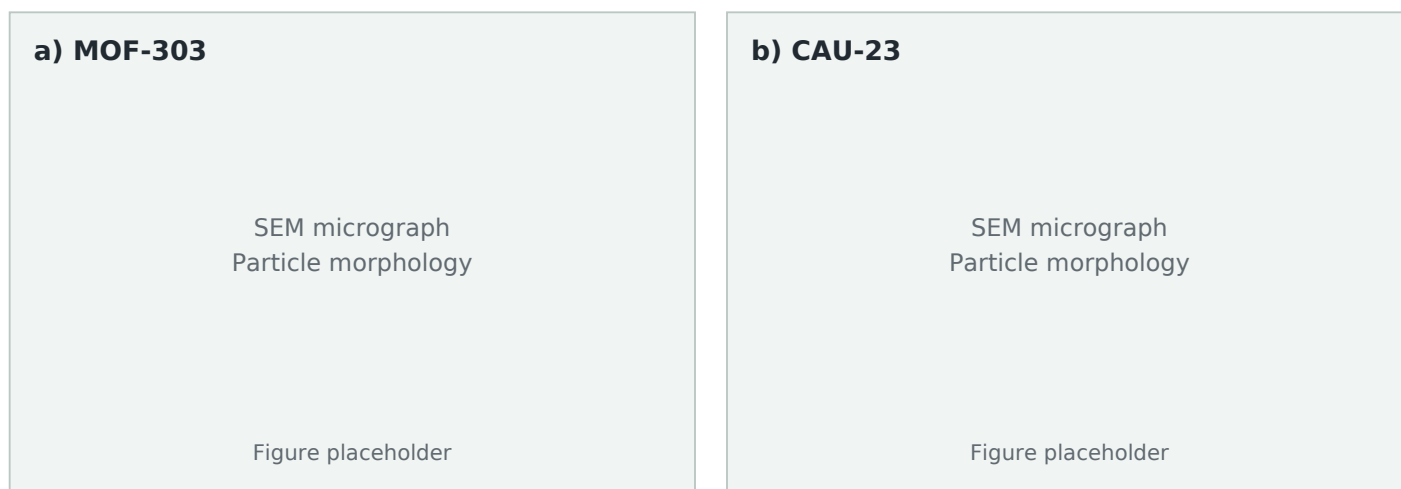


Figure S4. SEM images of (a) MOF-303 and (b) CAU-23. The panels are figure placeholders and contain no measured images.

References

- (1) Zheng, Z.; Hanikel, N.; Lyu, H.; Yaghi, O. M. Broadly Tunable Atmospheric Water Harvesting in Multivariate Metal-Organic Frameworks. *J. Am. Chem. Soc.* **2022**, *144*, 22669-22675. DOI: 10.1021/jacs.2c09756.
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- (4) Kim, H.; Cho, H. J.; Narayanan, S.; Yang, S.; Furukawa, H.; Schiffres, S.; Li, X.; Zhang, Y.-B.; Jiang, J.; Yaghi, O. M.; Wang, E. N. Characterization of Adsorption Enthalpy of Novel Water-Stable Zeolites and Metal-Organic Frameworks. *Sci. Rep.* **2016**, *6* (1), 19097.